

# TPC-396: Finite Signed-Law Interpolation and an Origin-Spread Transition

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## Abstract

TPC-395 transferred a law-dependent origin-spread split to a third coordinate-disjoint finite family. We now probe its mechanism by forming four exact linear combinations of the all-plus and alternating-index matrices at  $\lambda \in \{0, 1/3, 2/3, 1\}$  on a fourth fresh family. The complete panel has 96 rows and 16 cells. Twelve cells pass the one-percent origin-spread rule: all four normalizations pass at  $\lambda = 0, 1/3, 2/3$ , whereas all four fail at  $\lambda = 1$ , with maximal spreads between 0.0894220165 and 0.0940704384. Parent-relative holdout comparisons pass in all cells, but three scalar-normalized endpoint transfer cells exceed the fixed three-percent cap. The rational anchor proves the interpolation identity exactly. These are finite proxy observations and an obstruction-localization result, not a source-valid, asymptotic, arithmetic, Route-A/Route-B, or twin-prime theorem.

## 1 Question and scope

The preceding TPC-395 audit found that the all-plus and alternating-index finite matrices behave differently across origins, even after changing the affine family and the scalar normalization. The smallest next question is whether this split disappears continuously when the two endpoint matrices are mixed. We use a response-blind finite panel to locate a possible transition.

The objects below are deliberately proxy objects. In particular, a fractional coefficient in a matrix interpolation is not asserted to be an arithmetic character, a prime weight, or a new signed law. The claim firewall remains

$$\text{ARITHMETIC\_ADVANCE=NO,} \quad \text{FIXED\_POWER\_CREDIT=0,} \quad \text{FULL\_GATE\_B=OPEN.}$$

The official Session evaluator files are absent from this checkout; local proof and Bridge-B artifacts are fail-closed finite consistency evidence only.

## 2 Finite construction

For  $p \in (Q, 2Q]$ ,  $Q = 8192$ , put  $H = 66$  and define

$$K_p(u, v) = p(p/Q)^2 \frac{H^2}{H^2 + (u - v)^2} \left( \mathbf{1}_{p|u-v} - \frac{1}{p-1} \right) \\ \cdot \mathbf{1}_{u \neq v} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid v}.$$

Let  $G(u) = \sum_{p,v} K_p(u, v)^2$ . If the primes are ordered increasingly, the two endpoint matrices are

$$M_0 = \sum_p K_p, \quad M_1 = \sum_p (-1)^{\text{index}(p)} K_p.$$

For the four declared probes we use the exact finite identity

$$M_\lambda = (1 - \lambda)M_0 + \lambda M_1, \quad \lambda \in \{0, 1/3, 2/3, 1\}.$$

The fixed  $c = 3$  band retains pairs of the eight blocks whose block-index difference is at most three. We report the masked spectral diagnostic and, for an envelope check, the Schur diagnostic.

Table 1: Finite TPC-396 panel: 96 rows and 16 cells.

| normalization            | origin | cal | hold | transfer | max $R$      |
|--------------------------|--------|-----|------|----------|--------------|
| local diagonal           | 3/4    | 4/4 | 4/4  | 4/4      | 0.0894220165 |
| pooled train scalar      | 3/4    | 3/4 | 4/4  | 3/4      | 0.0940286260 |
| origin scalar            | 3/4    | 3/4 | 4/4  | 3/4      | 0.0940704384 |
| frozen train-1024 scalar | 3/4    | 3/4 | 4/4  | 3/4      | 0.0940286260 |

The current grid is

$$a_j = 6000001 + 401j \quad (0 \leq j < 41), \quad j \in \{0, 8, 16, 24, 32, 40\}.$$

The calibration origins are (6000001, 6003209, 6006417) and the holdouts are (6009625, 6012833, 6016041). Every origin has  $N = 1024$ . We apply local diagonal, pooled-calibration, current-origin, and first-calibration frozen normalizations. All roles, coefficients, and caps were fixed before current readout.

### 3 Statistics and certification

For a cell let  $S(o)$  denote its masked spectral value and define

$$R = \frac{\max_o S(o) - \min_o S(o)}{\text{mean}_o S(o)}.$$

The origin cap is  $R \leq 0.01$ . Parent-relative calibration and holdout errors compare the corresponding cohort means with the same-coefficient linear interpolation of the two frozen TPC-395 endpoint means. Their cap is 0.03; the within-family holdout/calibration error has the same cap.

The producer sums the shell in ascending order. An independent checker sums in descending order and reconstructs all rows without importing producer functions. Parent code and certificate hashes are verified first. The certificate is canonical JSON with a row digest and payload digest. A rational 13-point anchor at [6000001, 6000014] with  $Q = 8$  and shell {11, 13} checks positive geometry, endpoint symmetry, and all four interpolation identities. The mutation suite contains 28 contract mutations.

### 4 Results

Table 1 reports the normalization-level aggregates. The entry “origin” counts the four interpolation coefficients; “cal” and “transfer” use the fixed three-percent caps.

At the law level, ‘blend\_0’, ‘blend\_1\_3’, and ‘blend\_2\_3’ pass the origin cap for all four normalizations; ‘blend\_1’ passes none. Thus the origin-stable total is 12/16. The endpoint-1 origin spreads in local, pooled, origin, and frozen order are

$$0.089422016482946329, \quad 0.094028626026475742, \quad 0.094070438394687927, \quad 0.094028626026475617.$$

The largest parent-relative holdout error is 0.0033105775404086435, so every holdout comparison passes. Calibration passes are 4/4 locally and 3/4 for each scalar normalization. The largest within-family transfer errors are, in the same normalization order,

$$0.027691162656829471, \quad 0.030781025477051971, \quad 0.030792985412898766, \quad 0.030781025477051971.$$

The last three exceed 0.03 and are retained as failures. The finite spectral cap 0.64 fails on 24 rows (the six rows for ‘blend\_0’ in each normalization) and the Schur cap 0.83 fails on no row.

The exact anchor reports positive geometry and exact symmetry for all four probes. Since the fractional matrices are constructed from the two endpoint matrices, the rational identity is proved at the anchor; this does not promote the float64 finite observations to an asymptotic assertion.

## 5 Interpretation and next question

The strongest positive result is phase localization inside the declared finite panel: the three samples through  $\lambda = 2/3$  preserve origin stability and all parent-relative holdout levels remain close. The strongest obstruction is the endpoint  $\lambda = 1$ , where every normalization loses origin stability and three scalar transfer cells cross the fixed cap. The only justified transition statement is therefore the scoped finite observation that the tested endpoint behavior changes between  $\lambda = 2/3$  and  $\lambda = 1$ ; the interior interval  $(2/3, 1)$  remains untested, and no universal threshold follows.

The reusable structure is a hash-locked endpoint interface, exact rational interpolation, a fresh same-count calibration/holdout family, independent reverse-order replay, and an explicit mutation firewall. The next natural experiment is a response-blind replication with coefficients inside the untested interval, recorded as

ROUND2\_CLUE=TEST\_C1\_INTERPOLATION\_TRANSITION\_REPLICATION.

Source-valid origin uniformity, growing operator control, arithmetic  $L^2$ , Route closure, and the twin-prime endpoint remain open.

## Reproduction

The source, proof package, certificate, notes, and Bridge-B checker are stored under the TPC-396 project directory. The README gives ordinary and optimized commands for the producer, independent checker, stress test, and Bridge-B. The release requires ‘paper/main.pdf’ and ‘paper/paper.pdf’ to be byte-identical.